

Hex Color Model

The hex color model is a way of representing colors using hexadecimal values. In web design and development, it's commonly used to specify colors in HTML, CSS, and other related technologies.

In the hex color model, colors are represented by six characters: three pairs of hexadecimal digits, each pair representing the intensity of red, green, and blue (RGB) components of the color, respectively. Each pair can range from 00 to FF, where 00 is the lowest intensity (0) and FF is the highest intensity (255) for each color component.

- **#FF0000** represents pure red (255 for red, 0 for green and blue).
- **#00FF00** represents pure green (0 for red, 255 for green, 0 for blue).
- **#0000FF** represents pure blue (0 for red and green, 255 for blue).

RGB Color Model

RGB is an additive color model in which colors are created by combining red, green, and blue light in various intensities. Each color component (red, green, and blue) can have a value between 0 and 255, representing the intensity of that color. By adjusting the intensity of each color component, a wide range of colors can be created.

In CSS, colors can be specified using RGB values in several formats:

- **rgb(255, 0, 0)**: Specifies red with full intensity and no green or blue.
- **rgb(0, 255, 0)**: Specifies green with full intensity and no red or blue.
- **rgb(0, 0, 255)**: Specifies blue with full intensity and no red or green.
- **rgb(128, 128, 128)**: Specifies a shade of gray with equal intensities of red, green, and blue.

In addition to RGB, CSS also supports other color models such as HEX (Hexadecimal) and HSL (Hue, Saturation, Lightness). However, RGB remains the primary color model used in HTML and CSS for specifying colors.